SOFTWARE QUALITY ENGINEERING

Lab 7: Coverage Analysis

White-Box Testing with Coverage.py

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Course / Lab	Software Quality Engineering — Lab 7
Project	University Library Management System (auth.py)
Final test result	32 passed in 0.43 seconds
Final coverage	95% (above the lab's 90% checkpoint)

Purpose. This report records the baseline coverage, summarizes the gaps found, documents the added tests, and compares the final result with the baseline.

1. Baseline Coverage

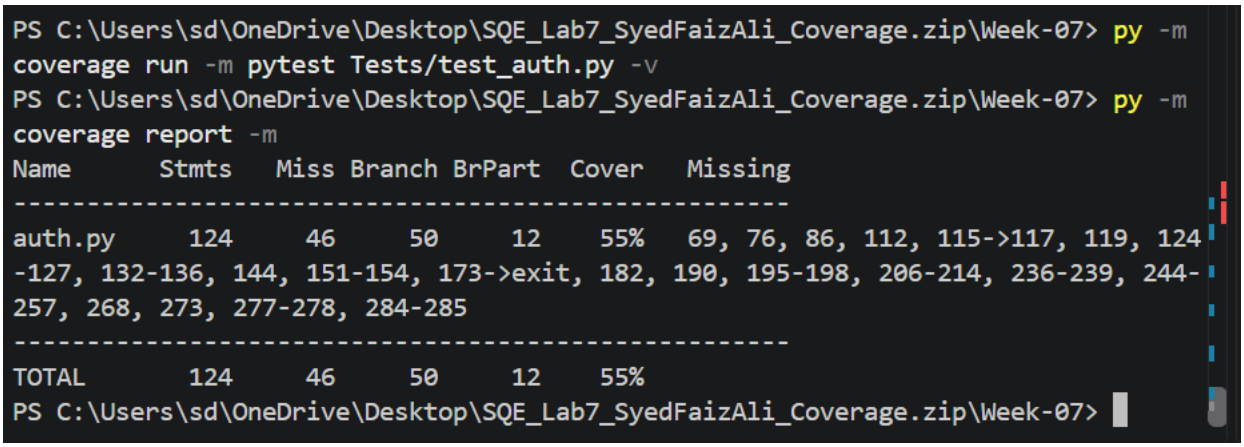
The original seven tests passed. The first recorded coverage run showed 55% total coverage.

Measure	Baseline value
Statements	124
Missed statements	46
Branches	50
Partial branches	12
Total coverage	55%
Tests passed	7

Baseline Missing list from the terminal:

69, 76, 86, 112, 115->117, 119, 124-127, 132-136, 144, 151-154, 173->exit, 182, 190, 195-198, 206-214, 236-239, 244-257, 268, 273, 277-278, 284-285.

Interpretation: the Missing column identifies statements or branch outcomes that were not exercised by the baseline tests. The line numbers reflect the version of auth.py used for this run.



2. Gap Analysis

The baseline results indicated that additional tests were needed for alternate and error paths, not just successful login. The following table maps the main behaviour gaps to targeted test ideas. The line groups are reported exactly as they appeared in the baseline output; because source line numbers can shift, function names should be confirmed in the annotated HTML report.

Baseline gap / area	Condition not covered	Test idea and expected result	Priority
69, 76, 86, 112	Alternate login and lockout outcomes	Test invalid input, unknown user, locked account, and failed-login threshold; assert the expected error and lock status.	High

115->117, 119, 124-127	Login/lockout branch outcomes	Exercise a valid login and boundary cases around failed attempts; assert success/failure and account state.	High
132-136, 144, 151-154	Account helper branches	Test locking, unlocking, unknown account handling, and resetting failed attempts; verify stored flags/counters.	Medium
173->exit, 182, 190	Session creation and validation paths	Test no session, valid session, and an expired session; assert the boolean result and session removal when expired.	High
195-198, 206-214	Reset-token marking and verification	Test a valid token, unknown token, expired token, and used token; expect True only for a valid unused token.	High
236-239, 244-257	Reset-email outcomes	Test unknown email, rate limit, token-generation exception, and SMTP ConnectionError; assert sent=False and the matching error.	High
268, 273, 277-278, 284-285	Remaining helper/error branches	Use targeted tests for the corresponding helper paths shown in htmlcov/auth_py.html; assert a specific return value or state change.	Medium

How to verify exact line-to-function mapping: open htmlcov/index.html, click auth.py, and use the red/yellow lines. The terminal output alone gives line numbers but not their function names, so the test ideas above are behaviour-focused rather than claiming an unverified exact mapping.

3. Tests Added and Final Results

The new tests targeted reset-token validation, lockout behaviour, login rejection paths, session handling, account helpers, and reset-email failure paths. The final test run recorded 32 passing tests.

Measure	Baseline	Final	Change
Tests passed	7	32	+25 passing tests
Statement count	124	124	No change
Missed statements	46	2	44 fewer missed statements
Branches	50	50	No change
Partial branches	12	7	5 fewer partial branches
Total coverage	55%	95%	+40 percentage points

Final report (terminal)

auth.py: 124 statements, 2 missed, 50 branches, 7 partial branches, 95% coverage. Missing: 126->exit, 134->exit, 153->exit, 173->exit, 197->exit, 213-214, 238->exit.

Coverage limitation

Code coverage shows which statements and branch outcomes ran; it does not prove that the tests checked the correct behaviour. A test can increase coverage even if it contains no meaningful assertion, so every test should verify an expected result.

Residual gaps and suggested tests

The final Missing list contains two missed statements (lines 213-214) plus several branch exits. The branch-exit entries indicate an alternate decision outcome that was not taken. Inspect those locations in the annotated HTML report before assigning exact function names. Add tests for the relevant alternate condition and assert the resulting return value or state change. The current result is already above the lab's stated 90% checkpoint.

```
PS C:\Users\sd\OneDrive\Desktop\SQE_Lab7_SyedFaizAli_Coverage.zip\Week-07> py -m
coverage run -m pytest Tests/ -v
PS C:\Users\sd\OneDrive\Desktop\SQE_Lab7_SyedFaizAli_Coverage.zip\Week-07> ^C
PS C:\Users\sd\OneDrive\Desktop\SQE_Lab7_SyedFaizAli_Coverage.zip\Week-07> py -m
coverage report -m
Name      Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------
auth.py     124      2     50      7    95%   126->exit, 134->exit, 153->exit, 17
3->exit, 197->exit, 213-214, 238->exit
-----------------------------------------------------
TOTAL       124      2     50      7    95%
```